

MPI Scatter and Gather Program Output

```
8.c
1  #include <stdio.h>
2  #include <mpi.h>
3
4  int main(int argc, char *argv[]) {
5      int rank, size, data[4], recv;
6
7      MPI_Init(&argc, &argv);
8
9      MPI_Comm_rank(MPI_COMM_WORLD, &rank);
10     MPI_Comm_size(MPI_COMM_WORLD, &size);
11
12     if (rank == 0) {
13         for (int i = 0; i < 4; i++)
14             data[i] = i + 1;
15     }
16
17     MPI_Scatter(data, 1, MPI_INT, &recv, 1, MPI_INT, 0, MPI_COMM_WORLD);
18
19     printf("Process %d received %d\n", rank, recv);
20
21     recv *= 2;
22
23     MPI_Gather(&recv, 1, MPI_INT, data, 1, MPI_INT, 0, MPI_COMM_WORLD);
24 }
```

Problems Output Debug Console Terminal Ports

PS C:\Users\Karthik R\Downloads\pc lab> gcc 8.c -o 8.exe -I"C:\Program Files (x86)\Microsoft SDKs\MPI\Include" -L"C:\Program Files (x86)\Microsoft SDKs\MPI\Lib\x64" -lmsmpi

PS C:\Users\Karthik R\Downloads\pc lab> mpiexec -n 4 .\8.exe

Process 3 received 4
Process 0 received 1
Gathered values: 2 4 6 8
Process 1 received 2
Process 2 received 3